CSE476 – Spring 2026
Mid-Term I – SET D

Name: ____________________ ASU ID: ____________________

Instructions

- Fill your Name and ASU ID above.
- There are 34 questions, the first question worth 1 point, and the remaining 33 questions worth 3 points each, totaling 100 points. This exam is graded out of 100 points.
- Select the one best answer for each question.
- You may use two cheat sheets, handwritten or typed, double-sided, and a calculator on this exam.
- Do not use AI, web or any other documents to answer the questions.
- By taking this exam, you agree to abide by the Academic Integrity policy established within the syllabus and any additional rules discussed in the class.

Single Answer Multiple Choice Questions

Select the one best answer for each question.

Q1. Which set is this?

(A) A
(B) B
(C) C
(D) D

Q2. Suppose two standard non-zero TF-IDF vectors have cosine similarity -0.2. What does this imply?

(A) The documents are strongly unrelated in topic
(B) They have negatively correlated term weights

(C) The similarity computation was implemented incorrectly

(D) The documents share no terms

Q3. Why do deeper networks require careful initialization?

(A) To control gradient scale across layers

(B) To decrease the vocabulary size of inputs

(C) To simplify the objective function

(D) To remove recurrence from the model

Q4. Stacking multiple linear layers without activations results in:

(A) Implicit regularization of weights

(B) Exponentially more expressive representations

(C) Increased ability to model hierarchical structure

(D) A model equivalent to a single linear transformation

Q5. Why do vanilla RNNs struggle with long-range dependencies?

(A) The hidden state has fixed dimensionality

(B) They reuse identical parameters at every time step

(C) Gradients decay or explode during backpropagation

(D) They produce outputs only at the final step

Q6. Why does the Markov assumption reduce sparsity in N-gram models?

(A) It removes independence assumptions

(B) It restricts conditioning context

(C) It increases vocabulary size

(D) It enforces smoothing

Q7. A two-layer neural network with step activations can represent the XOR function. However, when trained with gradient descent, it fails to learn XOR. What best explains this discrepancy?

(A) Step activations are insufficient to represent nonlinear decision boundaries

(B) The loss function is not convex for XOR

(C) The network requires more hidden units to separate XOR

(D) The activation provides no useful local gradient signal for weight updates

Q8. A neural language model is trained on the phrases ``small red car'' and ``large blue truck.'' During testing, it assigns meaningful probability mass to ``small blue car,'' which never appeared in training. What most directly enables this generalization?

(A) The model stores all observed word pairs explicitly

(B) Embeddings capture semantics that combine compositionally

(C) The softmax layer guarantees coverage of unseen sequences
(D) Training implicitly includes all possible word permutations

Q9. A TF-IDF retrieval system often misses semantically equivalent documents with synonyms. The primary limitation is:

(A) Bag-of-Words representation assumes term independence
(B) IDF overweights rare tokens
(C) Document length normalization distorts similarity
(D) Cosine similarity ignores magnitude differences

Q10. Increasing model size worsens test performance while improving training accuracy. This suggests:

(A) Optimization failure
(B) Underfitting
(C) Overfitting
(D) Over-regularization

Q11. When computing the loss at time t in an RNN, what differs from a feed-forward network?

(A) The RNN uses different weights at each step
(B) The RNN ignores its previous hidden state
(C) The RNN does not produce outputs at intermediate steps
(D) The loss depends on the hidden state from time $t-1$ and the same weight matrices are reused across all time steps

Q12. L2 regularization improves generalization primarily by:

(A) Forcing sparse parameter updates
(B) Encouraging smaller weight magnitudes
(C) Reducing the number of layers in the network
(D) Eliminating bias terms

Q13. A sigmoid neuron slows learning when its activation saturates because:

(A) Bias parameters stop updating
(B) Gradients become very small in extreme regions
(C) Weight magnitudes increase uncontrollably
(D) Its output becomes approximately linear

Q14. RNN models share parameters across time steps. Compared to a model with independent parameters at each step, weight reuse primarily:

(A) Allows parallel computation across sequence positions

(B) Eliminates vanishing gradients through parameter averaging
(C) Increases model capacity by expanding representational depth
(D) Reduces parameters and shares the same transformation

Q15. In RNN language modeling, the typical training loss is:

(A) Cross-entropy over the vocabulary
(B) Smooth L1 over embeddings
(C) Hinge loss over tags
(D) Binary cross-entropy per position

Q16. Suppose a document is tokenized at the sentence level (each entire sentence is one token). What impact does this have on vocabulary size and sequence length?

(A) Both vocabulary size and sequence length decrease
(B) Vocabulary size increases dramatically, and sequence length becomes very short
(C) Both vocabulary size and sequence length increase
(D) Vocabulary size decreases dramatically, and sequence length becomes extremely long

Q17. Why must an RNN be unrolled across time during training?

(A) To allocate separate parameters for each time step
(B) To represent recurrence as a time-indexed differentiable graph
(C) To remove weight sharing across sequence positions
(D) To increase hidden state dimensional capacity

Q18. LSTMs mitigate basic RNN issues mainly by:

(A) Freezing the recurrent matrix U
(B) Introducing gated operations
(C) Using convolution instead of recurrence
(D) Removing nonlinearity between layers

Q19. Which NLPar chitecture would be suitable for identifying the NER typef or` `Ben''i n ``Ben & Jerry's

(A) BiLSTM
(B) Word2Vec + Decoder RNN
(C) Bag-of-Words + Naive Bayes
(D) Hidden Markov Models

Q20. A 3-layer neural network with ReLU activations outputs only zeros for all inputs after initialization. What is the most likely cause?

(A) Cross-entropy loss misconfigured

(B) The final layer is using Softmax without temperature scaling
(C) Learning rate too high
(D) All pre-activation values are negative in the first layer

Q21. A 3-layer neural network with ReLU activations outputs a fixed, non-zero output for all inputs. What is the most likely cause?

(A) Dropout disabled
(B) The input data has not been normalized
(C) All weights are initialized to zero while biases are non-zero
(D) Learning rate too high

Q22. Compared to full-batch gradient descent, SGD introduces:

(A) Biased gradient estimates but lower variance
(B) Deterministic updates with faster convergence
(C) Unbiased gradient estimates with higher variance
(D) Smaller parameter magnitudes

Q23. Why does BPE reduce out-of-vocabulary issues?

(A) It removes infrequent words from the vocabulary
(B) It increases vocabulary size
(C) It decomposes rare words into subword units
(D) It converts tokens into fixed-length character blocks

Q24. A sequence model trained with teacher forcing performs well during training but poorly during inference. The most likely explanation is:

(A) Cross-entropy loss over-penalizes rare tokens during decoding
(B) The model experiences exposure bias due to distribution mismatch
(C) The hidden state dimensionality is insufficient for long sequences
(D) The optimizer fails to converge to a global minimum

Q25. In standard LSTMs, which component carries long-term summary information?

(A) Input embeddings
(B) Output logits
(C) Cell state c
(D) Hidden state h

Q26. Two non-zero TF--IDF document vectors have cosine similarity 0. What does this imply?

(A) Their Euclidean distance must be maximal

(B) The documents have different lengths

(C) The documents are unrelated in meaning

(D) The documents share no common terms

Q27. A tokenizer convertsa s entence intoa s equence like: $oncompressesthesentencei$. Whatbe ste xpla

(A) To enable the model to process text as numerical input based on a fixed vocabulary mapping

(B) To ensure every word has a single unchanging meaning

(C) To compress the sentence into a shorter representation

(D) To remove grammatical ambiguity before training

Q28. What is the probability of the sequence $H \to C \to H$? Given:

$$P(H \to H) = 0.7, \quad P(H \to C) = 0.3, \quad P(C \to H) = 0.4, \quad P(C \to C) = 0.6,$$

with initial probabilities $P(H) = 0.5, P(C) = 0.5$.

(A) 0.12

(B) 0.06

(C) 0.21

(D) 0.042

Q29. A language model achieves lower perplexity than another model but performs worse on question answering. The most principled explanation is:

(A) Lower perplexity always indicates overfitting to training data

(B) Perplexity ignores normalization across token positions

(C) Question answering requires larger vocabularies than modeling

(D) Perplexity reflects intrinsic likelihood rather than task utility

Q30. During training, a neural network's loss decreases for a few iterations, then begins oscillating widely without converging. Which explanation is most consistent with this behavior?

(A) The batch size is too small, increasing gradient variability

(B) The batch size is too large, reducing gradient noise

(C) The learning rate is too small to escape local minima

(D) The learning rate is too large for the local loss curvature

Q31. Consider a 4-word sequence $W = (w_1, w_2, w_3, w_4)$ with probabilities

$$P(w_1) = 0.5, \quad P(w_2 \mid w_1) = 0.4, \quad P(w_3 \mid w_1, w_2) = 0.2, \quad P(w_4 \mid w_1, w_2, w_3) = 0.1.$$

What is the perplexity of the sequence?

(A) 4.215

(B) 2.154

(C) 5.000

(D) 3.976

Q32. A system has two states, A and B, with the following transition probabilities:

- If in A: stays in A with probability 0.6 (and goes to B with probability 0.4)
- If in B: stays in B with probability 0.4 (and goes to A with probability 0.6)

Suppose the system starts in state B. As the number of transitions approaches infinity, what is the most accurate statement?

(A) The state probabilities converge to a fixed stationary distribution

(B) The state probabilities continue fluctuating without convergence

(C) The limiting distribution depends on the initial state

(D) The chain eventually remains permanently in one state

Q33. Int heW ord2Vec skip-gram model withn egative sampling, which oft hef ollowingb est describes and a context word c?

(A) Minimize the squared Euclidean distance between embeddings of all words in the same document

(B) Maximize $\log \sigma(v_c^\top v_w) + \sum_{i=1}^{k} \log \sigma(-v_{n_i}^\top v_w)$

(C) Maximize cosine similarity between all word pairs within a fixed window

(D) Maximize $\log P(c \mid w)$ using a full softmax over the vocabulary

Q34. What fundamentally distinguishes ELMo embeddings from Word2Vec?

(A) It represents words using character-level frequency features

(B) It learns embeddings solely from global co-occurrence statistics

(C) It produces context-dependent representations of words

(D) It assigns a single fixed vector to each word type

End of Paper